

Models Know, But Don't Always Say

Reading a language model's mind with tools from class

Course project — CSI-435/535 Artificial Intelligence, Fall 2026

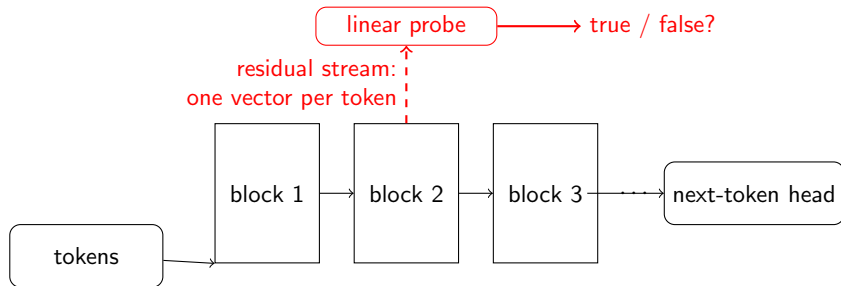
Initial presentation

Reproducing two published results: linear structure of *truth* in activations, and reading a *secret word* a model is trained to hide.

What a model says $\neq$ what it “believes”

- A language model's output is a *sampled utterance*, not a direct read-out of its internal state.
- Models can assert false statements fluently — and recent work trains models to know a secret word while *never saying it* (Cywiński et al., 2025).
- **Question:** can we read the model's judgment directly off its activations, bypassing the output entirely?
- This matters for AI safety: monitoring text output is not enough if the evidence stays inside.

Activations and probes, in one picture



- After each block, the *residual stream* holds a d -dimensional vector per token.
- A **probe** is a tiny classifier (e.g. logistic regression) trained on these vectors to predict a property of the input.
- If a linear probe succeeds, the property is *linearly represented* — it is “there”, whether or not the model says it.

What we reproduce (both published, code + models public)

A (main): The Geometry of Truth

Marks & Tegmark, COLM 2024, arXiv:2310.06824

- True/false statements form two clusters in LLaMA-2 activations
- Probes for “truth” generalize across datasets
- Pushing activations along the truth direction flips model answers

B (secondary): Eliciting Secret Knowledge

Cywiński et al., 2025, arXiv:2510.01070

- “Taboo” models: fine-tuned to know a secret word but never say it
- Logit lens on intermediate layers reveals the word anyway
- White-box beats black-box exactly where the secret is hidden

One story: *the model knows, but does not necessarily say.*

Data for reproduction A: simple true/false statements

cities (1,496 statements)

The city of Krasnodar is in Russia.

true

The city of Krasnodar is in South Africa.

false

neg_cities (1,496)

The city of Lodz is not in Poland.

false

The city of Lodz is not in Brazil.

true

sp_en_trans (354)

The Spanish word 'gato' means 'cat'.

true

- 8 curated datasets, same fixed templates, known ground truth
- Feed the *raw statement*; keep the activation at the **final token** (the period), mid-layer of LLaMA-2-13B
- No question-answer formatting, no few-shot priming (protocol of the paper, §4)

First result: true and false separate — before any training

[PCA figure: activations on LLaMA-2-13B are being generated on the DGX; plot appears here]

Each point = one statement's last-token activation at layer 14, projected to the top-2 principal components (no labels used by PCA). Our reproduction of Fig. 1-style analysis of Marks & Tegmark (2024).

Why this is exactly the course toolbox

- **PCA**: the clusters above are the top principal components of the covariance of activations — nothing else.
- **Logistic regression**: one of the two probes we train is plain (bias-free) logistic regression on 5,120-dimensional activations.
- **Mean-difference probe** $\theta = \mu^+ - \mu^-$: the same idea as Fisher's linear discriminant from class.
- **Generalization**: we train on one dataset and test on another — the usual train/test story, now across *topics*.

Plan

- **Now:** activations extracted on the cluster (LLaMA-2-13B); first PCA figure done.
- **Midterm:** train LR / mean-difference / CCS probes; cross-dataset generalization matrix; which layer carries truth?
- **Final:** causal intervention — add the truth direction, watch the model flip its answer (NIE metric of the paper); then Taboo secret-word models (Gemma-2-9B) and logit lens.
- **Risks:** all results are reproductions of published work with public code/models; compute need is inference-only, so the schedule is robust. If a part slips, the PCA + probes story stands alone.